Kelvin Andrei Cahuana Condori

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Summary

Computer Science undergraduate at the University of Engineering and Technology (UTEC) with strong research interests in quantum computing, quantum learning theory, deep learning, and post-quantum cryptography. I have participated in quantum computing research projects and hackathons using Qiskit, PennyLane, and Cirq, including simulations of quantum scrambling and the development of hybrid quantum-classical models for climate data analysis and prediction. I am seeking opportunities to join a research group in academia or industry, where I can contribute to ongoing research, collaborate on applied projects, further develop my research skills, and pursue a long-term career in quantum technologies.

Research Interests

Quantum Learning Theory (QLT): learnability, sample complexity, quantum state discrimination, shadow tomography, and quantum property testing.

Quantum Optimization: quantum optimization algorithms, hybrid quantum-classical methods, combinatorial optimization, and variational optimization techniques.

Quantum Information: classical shadows, quantum state tomography, efficient observable estimation, and quantum system characterization.

Quantum Algorithms: quantum advantage, dequantization, quantum learning algorithms, and quantum information processing.

Post-Quantum Cryptography: post-quantum cryptographic algorithms, lattice-based cryptography, cryptographic security, and quantum-resistant cryptographic schemes.

Technical Skills

Programming Languages: Python, C++, C, C#, JavaScript, TypeScript, Rust, and Bash.

Quantum Computing: Qiskit, PennyLane, Cirq, D-Wave Ocean SDK, variational quantum algorithms (VQAs), and quantum annealing.

Scientific Computing: NumPy, SciPy, Matplotlib, Jupyter Notebook, and data analysis.

Machine Learning & Deep Learning: Scikit-learn, PyTorch, generative models, and hybrid quantum-classical models.

Development Tools: LaTeX, Git, GitHub, Linux, Jupyter Notebook, and Visual Studio Code.

Research Skills: Critical analysis of scientific literature and reproducible research practices.

Languages: Spanish (native) and English (intermediate).

Selected Projects in Quantum Computing and Related Fields

Speaker on *Quantum vs. Classical Advantage* October 2026
IBM Qiskit Fall Fest Peru 2026 - Stage 2 *Quantum Computing, Quantum Advantage, Dequantization*

- Delivered a Spanish-language talk titled “Quantum vs. Classical Advantage: ¿Un logro alcanzado o un objetivo en movimiento?” as part of Stage 2 of IBM Qiskit Fall Fest Peru 2026.
- Discussed the historical development of quantum advantage, from the concept of quantum supremacy to the challenges of demonstrating practical quantum computational advantages.
- Explored the role of the Noisy Intermediate-Scale Quantum (NISQ) era and fault tolerance in the evolution of quantum computing.
- Examined dequantization and the challenges of establishing genuine quantum advantage over classical approaches.
- Event details: IBM Qiskit Fall Fest Peru 2026 - Stage 2.

Lecturer on *Quantum Entanglement and Quantum Protocols* September 2026
IBM Qiskit Fall Fest Peru 2026 - Stage 1 *Quantum Entanglement, Quantum Information, Quantum Protocols*

- Delivered a lecture on *quantum entanglement* and its role in quantum information processing and quantum communication protocols.
- Introduced fundamental concepts of quantum entanglement and discussed their applications in quantum protocols.
- Contributed to the academic program of IBM Qiskit Fall Fest Peru 2026 by delivering an educational session on quantum information.
- Event details: Entanglement and Quantum Protocols - Qiskit Fall Fest Peru.

Analysis of the Dequantization of Variational Quantum Algorithms Using Amplitude Encoding 2026 - Present
Undergraduate Research Thesis *Dequantization, QML, Quantum Algorithms*

- Investigating the *dequantization* of *variational quantum algorithms* used in *quantum machine learning*, with particular emphasis on algorithms employing *amplitude encoding*.
- Studying when variational quantum models can be efficiently simulated by classical algorithms and identifying the assumptions that enable such classical simulations.
- Analyzing the role of *amplitude encoding*, classical data access, and data-loading assumptions in determining whether a quantum advantage can be maintained or reproduced classically.
- Exploring connections between *dequantization*, *quantum advantage*, and classical simulation techniques in the context of variational quantum algorithms.
- Reviewing theoretical results on quantum machine learning dequantization and quantum advantage, including work by Ewin Tang, Scott Aaronson, and other researchers studying classical algorithms for quantum-inspired and quantum machine learning methods.

Q-ENSO: Hybrid Quantum-Classical Drought Predictor October 2025
Second Place, Quantum Hackathon LATAM 2025 *PennyLane, Quantum Machine Learning, VQAs*

- Optimized the **SQUARE Mamba** model for drought forecasting in Paso Severino, Uruguay.
- Developed a hybrid quantum-classical pipeline for training and evaluating climate forecasting models.
- Integrated PennyLane and qBraid for the simulation and execution of the model's quantum components.
- Evaluated the robustness of the model through quantum noise simulations and their impact on predictive performance.
- Applied quantum computing techniques to climate forecasting and water resource management problems.
- Awarded Second Place at Quantum Hackathon LATAM 2025 among participating teams from across Latin America.
- Repository: github.com/andrewkc/Q-ENSO

Quantum Scrambling Verification Simulator May 2026 - August 2026
Final Project, Introduction to Quantum Information *Qiskit, Quantum Information, Quantum Scrambling*

- Implemented in Qiskit the *quantum scrambling* verification protocol presented in **Verified Quantum Information Scrambling** (Landsman et al., 2019), based on the quantum teleportation proposal by Yoshida and Yao.
- Simulated quantum circuits involving EPR pairs, Bell measurements, and quantum teleportation protocols.
- Investigated the relationship between *out-of-time-ordered correlators* (OTOCs), quantum scrambling, and decoherence processes.
- Implemented three-qubit unitary scramblers and analyzed quantum information scrambling in multipartite quantum systems.
- Compared direct OTOC computation with the probabilistic and deterministic teleportation protocols reported in the literature, including the Grover-based variant that replaces Bell-measurement post-selection.
- Evaluated the metrics P_ψ (Bell measurement success probability), F_ψ (teleportation fidelity), and N (noise factor) to distinguish genuine quantum scrambling from decoherence.
- Repository: github.com/andrewkc/quantum-scrambling

Education

University of Engineering and Technology (UTEC) Lima, Peru
B.S. in Computer Science *August 2021 - Agosto 2027*

Activities & Societies: *Advanced Mathematics Club, Competitive Programming Club, Free Software Club*

Experience

Co-Founder August 2026 - Present
QPeru *Peruvian Community for Quantum Computing and Related Technologies*

- Co-founded QPeru, a Peruvian community dedicated to quantum computing, quantum information, and related technologies.
- QPeru is an official prospective QCousin of QWorld, reflecting its commitment to expanding quantum computing education and outreach in Peru. LinkedIn.
- Contribute to educational, research, and community-building initiatives to strengthen Peru's quantum computing ecosystem.
- Collaborate with students, researchers, and professionals to promote knowledge exchange, technical training, and research opportunities in quantum technologies.
- Organize and deliver workshops on quantum computing and quantum technologies, fostering academic engagement and community growth.
- **Qiskit Fall Fest 2026:** Co-organize this event with IEEE Computer Society PUCP and the PUCP Physics Society to promote quantum computing education and hands-on learning.

• **IBM Qiskit Fall Fest Peru 2026 - Stage 1:** Co-developed the syllabus and co-organized the sessions for a 10-day introductory workshop on quantum computing and quantum physics, in collaboration with the PUCP Physics Students Association. Event announcement.

• **IBM Qiskit Fall Fest Peru 2026 - Stage 2:** I coordinated with Sociedad Inducontrol Ingeniería S.A.C. to organize a technical talk and live demonstration of the *SpinQ Gemini Mini*, a two-qubit quantum computer based on Nuclear Magnetic Resonance (NMR) technology. Additionally, I personally handled the invitation and coordination of speakers for the IBM Quantum University Tour at UTEC on October 23, 2026.

• Website: qperu.net

Events and Projects Team August 2026 - Present
IEEE Computer Society UTEC *Lima, Peru*

• Support the planning and execution of technical events and projects.

• Promote quantum computing activities and initiatives within the university community.

Research Team August 2026 - Present
IEEE Computer Society PUCP *Lima, Peru*

• Contribute to research initiatives in quantum computing and related fields.

• Support research activities and knowledge sharing within the student community.

Open-Source Contributions

Qiskit Documentation October 2026
Typographical correction in Spanish learning materials *IBM Quantum*

• Reported a typographical inconsistency in the Spanish CHSH Game lesson, proposing the use of em dashes to correctly delimit a parenthetical phrase.

• Issue: GitHub Issue

Honors and Awards

117th Place out of 17,000+ Participants - QHack 2026 October 2026
PennyLane by Xanadu, International Quantum Computing Hackathon

Selected Participant - First Quantum Computing School (Quantum Initiative) September 2026
Universidad Nacional de Río Negro (UNRN) and Universidad Nacional de Hurlingham (UNAHUR)

Edwin Catmull Award - Best Computer Graphics Project (2026-I) July 2026
Faculty of Computing, University of Engineering and Technology (UTEC)

Selected for the 2026 Hackathon - Quantum Computing for Water Challenges June 2026
Benemérita Universidad Autónoma de Puebla (BUAP), Open Quantum Institute

Second Place - Quantum Hackathon LATAM 2025: Quantum for Climate October 2025
University of Montevideo, Open Quantum Institute (CERN)

Panelist, Quantum AI Summit 2025 December 2025
QuantumHub Peru, Center for Scientific Outreach, Culture and Technological Development (CEDDITEC)

Recipient of the Beca 18 National Scholarship May 2021
National Scholarship and Educational Loan Program (PRONABEC)

Certifications

Qiskit Global Summer School 2026 Excellence Badge July 2026
IBM Quantum

Selected Topics in Quantum Science and Technologies (QCourse 581) February 2026 - June 2026
QWorld and University of Latvia

QWorld OQI Pre-Hackathon Training Programme 2026 (QCobalt) June 2026
Open Quantum Institute and QWorld

QWorld QSilver Workshop February 2026
QEgypt and QTunisia

QWorld OQI Hackathon QC Certification Courses 2025 (QBronze and QNickel) August 2025
Open Quantum Institute and QWorld

Quantum Machine Learning Workshop October 2025
Fundacja Quantum AI and QPoland

Fundamentals of Quantum Information January 2026
IBM Quantum Platform

Other Interests

Geometric algebra, abstract algebra, *Langlands program*, functional analysis, computational complexity theory, theoretical computer science, competitive programming, *deep learning*, *statistical learning theory*, generative models, quantum mechanics, *quantum error correction*, analytic philosophy, music theory, music production, musicology, and classical and contemporary literature (*dark academia*).